BASIC NETWORKING COMMANDS IN LINUX AND WINDOW OS

Ex.No:1
16/7/25

AIM:

To study and execute basic networking commands in windows and linux OS for viewing network config, testing connectivity, checking route information, monitoring network status and troubleshoot network related issues.

COMMANDS:

1) **IP CONFIG:**

Displays network configuration.

- WINDOWS COMMAND :- ipconfig
- LINUX COMMAND :- ip addr (or) ifconfig

2) **PING:**

Test connectivity to a remote host.

- WINDOWS COMMAND:- ping google.com
- LINUX COMMAND - ping google.com

3) **TRACE ROUTE:**

Display the path packets take to reach a destination

- WINDOWS COMMAND - tracert google.com
- LINUX COMMAND - traceroute google.com

4) **NSLOOKUP:**

Resolves domain names to IP address.

- WINDOWS COMMAND - nslookup google.com
- LINUX COMMAND - nslookup google.com

5) NETSTAT:

Displays active network connections and listening ports.

WINDOWS COMMAND :- netstat / netstat -a

LINUX COMMAND - netstat -a / ss -a

6) ARP:

Displays the ARP cache

WINDOWS COMMANDS :- arp -a

LINUX COMMANDS - arp -a / ip neigh

7) HOSTNAME:

Display the system hostname

WINDOWS COMMAND :- hostname

LINUX COMMAND - hostname.

8) route:

Displays or modifies the routing table.

WINDOWS COMMAND - route print

LINUX COMMAND - ip route

9) MAC address:

Display MAC address of the computer:

WINDOWS COMMAND - getmac

LINUX COMMAND - ip link

RESULT:

Thus basic networking commands are executed in both windows CMD and linux terminal successfully.

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AIM:

To study and execut

```
[sanjeetha@localhost ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fef2:96fd  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:f2:96:fd  txqueuelen 1000  (Ethernet)
        RX packets 26  bytes 3188 (3.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 69  bytes 7363 (7.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 18  bytes 2112 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 18  bytes 2112 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[sanjeetha@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[sanjeetha@localhost ~]$ hostname
localhost.localdomain
[sanjeetha@localhost ~]$ nslookup google.com
Server:         10.252.249.176
Address:        10.252.249.176#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.205.14
Name:   google.com
Address: 2404:6800:4007:801::200e

[sanjeetha@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
```

```
Address: 2404:6800:4007:801::200e

[sanjeetha@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp        0      0 localhost.localdo:57058 mirror.twds.com.t:https TIME_WAIT
tcp        0      0 localhost.localdo:52042 ftp.yz.yamagata-u.:http TIME_WAIT
tcp        0      0 localhost.localdo:57086 mirror.twds.com.t:https TIME_WAIT
tcp        0      0 localhost.localdo:57074 mirror.twds.com.t:https TIME_WAIT
tcp        0      0 localhost.localdo:50946 250621808.mrs.cdn:https ESTABLISHED
tcp        0      0 localhost.localdo:52052 ftp.yz.yamagata-u.:http TIME_WAIT
tcp        0      0 localhost.localdo:54010 ftp.yz.yamagata-u.:http TIME_WAIT
tcp        0      0 localhost.localdo:52066 ftp.yz.yamagata-u.:http TIME_WAIT
tcp        0      0 localhost.localdo:53984 ftp.yz.yamagata-u.:http TIME_WAIT
tcp        0      0 localhost.localdo:37922 mirror.twds.com.t:https TIME_WAIT
tcp        0      0 localhost.localdo:54000 ftp.yz.yamagata-u.:http TIME_WAIT
tcp        0      0 localhost.localdo:51550 ftp.yz.yamagata-u.:http TIME_WAIT
tcp6       0      0 localhost:ipp           [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
udp        0      0 localhost.locald:bootpc _gateway:bootps         ESTABLISHED
udp        0      0 0.0.0.0:mdns            0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp6       0      0 [::]:mdns               [::]:*
udp6       0      0 localhost:323           [::]:*
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  2      [ ACC ]     STREAM     LISTENING     29122    @/tmp/.ICE-unix/2498
unix  2      [ ACC ]     STREAM     LISTENING     20872    /run/systemd/private
cket
unix  2      [ ACC ]     SEQPACKET  LISTENING     20887    /run/lvm/lvmpolld.so
p
unix  2      [ ACC ]     SEQPACKET  LISTENING     20889    /run/systemd/coredum
unix  2      [ ACC ]     STREAM     LISTENING     20891    /run/udev/control
unix  2      [ ACC ]     STREAM     LISTENING     29379    /tmp/.X11-unix/X0
                                                 29381    /tmp/.X11-unix/X1
```

4) NSLOOKUP:

```
MTU    OMTU Security
Proto  Destination       Source              State     Channel
[sanjeetha@localhost ~]$ arp -a
_gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3
[sanjeetha@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAUL
T group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP m
ode DEFAULT group default qlen 1000
    link/ether 08:00:27:f2:06:fd brd ff:ff:ff:ff:ff:ff
[sanjeetha@localhost ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 10.252.249.176
[sanjeetha@localhost ~]$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAUL
T group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
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ode DEFAULT group default qlen 1000
    link/ether 08:00:27:f2:06:fd brd ff:ff:ff:ff:ff:ff
[sanjeetha@localhost ~]$ ss -tuln
Netid  State    Recv-Q   Send-Q     Local Address:Port      Peer Address:Port

udp    UNCONN   0        0              0.0.0.0:5353           0.0.0.0:*

udp    UNCONN   0        0            127.0.0.1:323            0.0.0.0:*

udp    UNCONN   0        0                [::]:5353              [::]:*

udp    UNCONN   0        0               [::1]:323               [::]:*

tcp    LISTEN   0        4096         127.0.0.1:631            0.0.0.0:*

tcp    LISTEN   0        128            0.0.0.0:22             0.0.0.0:*

tcp    LISTEN   0        4096             [::1]:631              [::]:*
```

```
    link/ether 08:00:27:f2:06:fd brd ff:ff:ff:ff:ff:ff
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udp    UNCONN   0        0            127.0.0.1:323            0.0.0.0:*

udp    UNCONN   0        0                [::]:5353              [::]:*

udp    UNCONN   0        0               [::1]:323               [::]:*

tcp    LISTEN   0        4096         127.0.0.1:631            0.0.0.0:*

tcp    LISTEN   0        128            0.0.0.0:22             0.0.0.0:*

tcp    LISTEN   0        4096             [::1]:631              [::]:*

tcp    LISTEN   0        128              [::]:22                [::]:*

[sanjeetha@localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
[sanjeetha@localhost ~]$
```